

Total No. of Questions : 6]

SEAT No. :

P5043

[Total No. of Pages : 2

TE/Insem - 541

T.E. (I.T.) (Semester - I)

Computer Network Technology

INFORMATION TECHNOLOGY

(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4. Q5 or Q.6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) List out and explain features of OSPF that makes it superior to RIP. [6]
- b) Is static and Dynamic IP address allocation supported by DHCP protocol? Explain the state transition diagram of DHCP client and explain in brief. [4]

OR

- Q2)** a) An organization is granted the block 192.168.10.0/27. Design the network and i) Find how many subnets? ii) Find how many hosts per subnet? iii) Subnet mask iv) What are the valid subnets? v) What's the broadcast address for last subnet? vi) What is the range of valid hosts in last subnet?[6]

- b) What is the use of ARP and RARP? Explain ARP operations. [4]

- Q3)** a) What is the purpose of Leaky bucket and token bucket algorithms? Describe working of Leaky bucket algorithm with reference to CBR, VBR and bursty traffic. [6]

- b) List the fields in TCP header that are not part of UDP header. Give the reasons of each missing field. [4]

P.T.O.

OR

Q4) a) What is socket? Explain various socket primitives used in client server interaction. [6]

b) Explain how flow control is different than congestion control in TCP?. [4]

Q5) a) What is FTP? Where and when it is used? Why it faster? Explain at least 3 user commands used in FTP. [6]

b) What is MIME? Explain the need of MIME with suitable example. [4]

OR

Q6) a) What are the query resolution techniques in DNS? Explain any one of them. [6]

b) Is HTTP connection oriented or connection less protocol? Differentiate between persistent and non-persistent HTTP connection. [4]

